GEOTECHNICAL ENGINEERING

UNIT - V : SHEAR STRENGTH OF SOILS

Course Code: 23A01503T

Syllabus: SHEAR STRENGTH OF SOILS: Importance of shear strength - Mohr's-Coulomb failure theories - Types of laboratory tests for strength parameters - Strength tests based on drainage conditions - Critical void ratio - Liquefaction.

1. Importance of Shear Strength

Shear strength of a soil is the maximum resistance that the soil offers to shear stresses (sliding of one part of the soil mass over the other) before it fails. Soil fails in shear, never in compression or tension - so the strength of soil means its shear strength.

- Bearing capacity of foundations - the soil below the footing fails by shearing.
- Stability of slopes - embankments, earth dams, cuts, natural hill slopes.
- Earth pressure on retaining walls, sheet piles, bridge abutments, braced cuts.
- Stability of pavements, subgrade and the design of tunnels, pipelines and underground structures.
- Safety of excavations, the stability of the bases of the pile foundations and the skin friction of piles.

Sources of shear strength in a soil

- Cohesion (c) - the attraction between the particles of fine-grained soils (clay), due to electro-chemical bonds and adsorbed water, and cementing agents. It is independent of the normal stress.
- Internal friction (φ) - the frictional resistance between the grains: (i) sliding friction at contacts, (ii) interlocking of particles (must dilate to move over each other), (iii) rolling friction. It is proportional to the normal stress on the plane.

Clean sand and gravel: $c = 0$ (cohesionless, "frictional" soil). Saturated clay in undrained condition: $\varphi = 0$ (purely cohesive soil). Most natural soils (silty clay, clayey sand) are c - φ soils.

2. Stresses on a Plane - Mohr's Circle

In a stressed soil element, three mutually perpendicular planes have no shear stress on them - these are the principal planes, and the normal stresses on them are principal stresses: major σ_1, intermediate σ_2, minor σ_3. For two-dimensional problems σ_2 is ignored.

Analytical expressions

On a plane inclined at θ to the major principal plane (angle θ is measured from the plane carrying σ_1):

$$\sigma_n = (\sigma_1 + \sigma_3)/2 + (\sigma_1 - \sigma_3)/2 \cos 2\theta$$

$$\tau = (\sigma_1 - \sigma_3)/2 \sin 2\theta$$

- Maximum shear stress $\tau_{max} = (\sigma_1 - \sigma_3)/2$ at $\theta = 45^\circ$ (radius of the circle).

Mohr's circle (graphical)

- Take σ on the x - axis and τ on the y - axis (compression positive in soil mechanics).
- Mark σ_3 and σ_1 on the axis; the circle having the diameter $(\sigma_1 - \sigma_3)$ is the Mohr's circle. Centre at $(\sigma_1 + \sigma_3)/2$, radius $(\sigma_1 - \sigma_3)/2$.
- Every point on the circle gives the (σ, τ) on one plane. A plane at θ is represented by the point at an angle 2θ from the σ_1 end, measured at the centre.
- The pole (origin of planes) method: draw through the point of σ_3 a line parallel to the plane of σ_3, the intersection with the circle is the pole; the lines from the pole to any point of the circle give the orientation of the plane.

Example 1

Major and minor principal stresses at a point are 400 kN/m^2 and 100 kN/m^2. Find the normal and shear stress on a plane inclined at 30° to the major principal plane.

- $\sigma_n = (400+100)/2 + (400-100)/2 \cos 60^\circ = 250 + 150(0.5) = 325$ kN/m^2
- $\tau = (400-100)/2 \sin 60^\circ = 150 \times 0.866 = 129.9$ kN/m^2; $\tau_{max} = 150$ kN/m^2 (at 45°).

3. Mohr - Coulomb Failure Theory

Coulomb (1776) proposed that the shear strength of soil on a plane is a linear function of the normal stress on it:

$$\tau_f = c + \sigma \tan \varphi$$

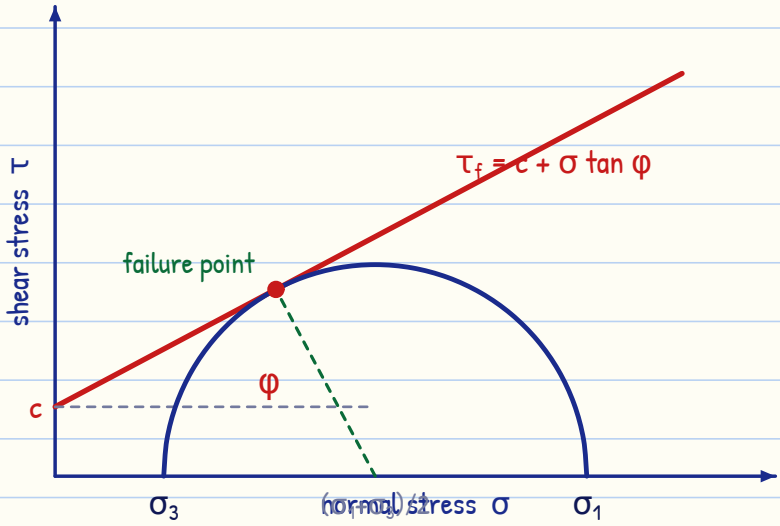
where τ_f = shear strength, c = cohesion intercept, φ = angle of shearing resistance (angle

of internal friction), σ = normal stress on the failure plane.

- Terzaghi's effective stress principle - the shear strength is controlled by the effective normal stress, hence:

$$\tau_f = c' + (\sigma - u) \tan \varphi' = c' + \sigma' \tan \varphi'$$

- Mohr (1900) postulated that a material fails when the shear stress on the failure plane reaches a function of the normal stress; the failure envelope is a curve. For the stress range met in practice the envelope is approximated by a straight line (Coulomb), hence the Mohr - Coulomb failure criterion.



Failure plane makes
$\theta = 45° + \varphi/2$
with the major principal plane;
the angle $2\theta = 90° + \varphi$
is measured at the centre.

Circle touches the envelope at failure; if it stays below the envelope the soil is safe.

Conclusions from the Mohr - Coulomb diagram

- The failure envelope is the tangent to all the Mohr's circles at failure. If the circle lies below the envelope, the soil is stable; if it touches, it is at failure; a circle above the envelope is impossible (the soil would already have failed).
- Failure takes place on the plane at $\theta_f = 45^\circ + \varphi/2$ to the major principal plane (not on the plane of maximum shear stress, 45°, as the friction acts also).
- The normal and shear stresses on the failure plane: $\sigma_f = (\sigma_1+\sigma_3)/2 + (\sigma_1-\sigma_3)/2 \cos 2\theta_f = (\sigma_1+\sigma_3)/2 - (\sigma_1-\sigma_3)/2 \sin \varphi$, $\tau_f = (\sigma_1-\sigma_3)/2 \cos \varphi$.

Relation between the principal stresses at failure

From the right-angled triangle of the Mohr's circle (centre, tangent point and the point where the envelope meets the σ axis at -c cot φ):

$$\sin \varphi = (\sigma_1 - \sigma_3) / (\sigma_1 + \sigma_3 + 2c \cot \varphi)$$

$$\sigma_1 = \sigma_3 \tan^2(45^\circ + \varphi/2) + 2c \tan(45^\circ + \varphi/2)$$

$$\sigma_3 = \sigma_1 \tan^2(45^\circ - \varphi/2) - 2c \tan(45^\circ - \varphi/2)$$

- $N_\varphi = \tan^2(45^\circ + \varphi/2)$ is called the flow value. For sands ($c = 0$): $\sigma_1/\sigma_3 = N_\varphi$, $\sin\varphi = (\sigma_1 - \sigma_3)/(\sigma_1 + \sigma_3)$.
- For purely cohesive soils ($\varphi = 0$): $\sigma_1 = \sigma_3 + 2c$, i.e. $c = (\sigma_1 - \sigma_3)/2 = \tau_{max}$ and the failure plane is at 45°.

Example 2

(a) A dry sand with $\varphi = 30^\circ$ is tested in triaxial with $\sigma_3 = 100$ kN/m^2. Find σ_1 at failure.

- $\sigma_1 = \sigma_3 \tan^2(60^\circ) = 100 \times 3 = 300$ kN/m^2; the plane of failure is at $45^\circ + 15^\circ = 60^\circ$ with the major principal plane.

(b) A clay with $c = 20$ kN/m^2, $\varphi = 20^\circ$, $\sigma_3 = 100$ kN/m^2:

- $\tan(55^\circ) = 1.428$, $\tan^2(55^\circ) = 2.039$; $\sigma_1 = 100(2.039) + 2(20)(1.428) = 203.9 + 57.1 = 261$ kN/m^2.

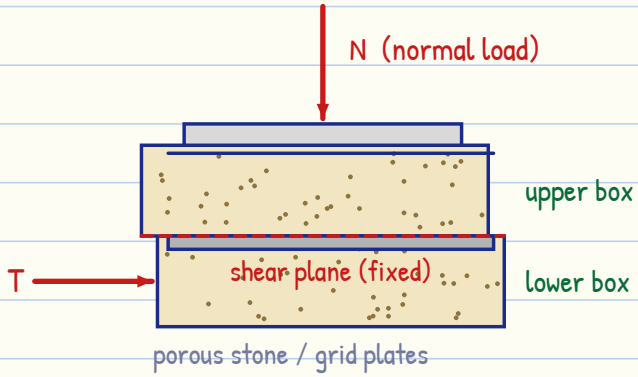
4. Types of Laboratory Tests for Strength Parameters

The shear strength parameters c and φ are determined from:

Test	Type of stress	Used for
Direct shear test (shear box)	failure plane is forced to be horizontal	sands, gravels, quick test on all soils
Triaxial compression test	all round pressure and axial load	all soils, any drainage condition (most versatile)
Unconfined compression test (UCC)	$\sigma_3 = 0$, special case of triaxial	saturated cohesive soils (clay), only c_u
Vane shear test	torque on the vanes	soft clays and silts (lab and field)

- Other tests: simple shear, ring shear, plane strain and torsion tests (research), and in-situ tests such as SPT, CPT and field vane.
- Selection of the test is made on the basis of the soil type, the available time and the field drainage conditions.

5. Direct Shear Test (IS 2720 Part 13)



Box 60 x 60 mm (or 100 x 100 mm)
Rate of shearing 1.25 mm/min
(sand 0.2 - 1 mm/min)

$\tau = T / A$, $\sigma = N / A$
Plot τ_f against σ at three values of N -> envelope:
slope = φ, intercept = c

Procedure

- The soil sample (sand compacted to the required density, or clay at the required water content) is placed in the shear box between two porous stones with grid plates (for drained test). The box is split horizontally into two halves, so the shear plane is predetermined.
- A vertical normal load N is applied through the loading pad; then the lower box is moved at a constant rate (strain controlled), the upper box being held by the proving ring, which measures the shear force T. Horizontal and vertical dial gauges record the displacement (and the volume change).
- The test is carried out on at least three identical specimens under different normal stresses (say 50, 100, 150 kPa), and the maximum shear stress (peak), $\tau_f = T_{max}/A$ is found for each. A (area) is corrected for the reduction in the shear plane as shearing proceeds (optional).
- Plot τ_f against σ: a straight line; slope = φ, intercept on the τ axis = c.

Types of direct shear test

- Undrained (quick) test - shearing begins immediately after applying the load (about 1 - 2 min), no time for drainage; Consolidated undrained - shear after consolidation under N, with the drainage closed; Drained (slow) test - the soil is allowed to consolidate under N, then sheared slowly (so that the pore pressure is nil): φ_d, c_d.
- For cohesionless soils the drained test is quick since drainage is free. For clay, the shearing time is $t_f = 12.7\ t_{100}$ (t_{100} from the consolidation) - several days.

Advantages and disadvantages

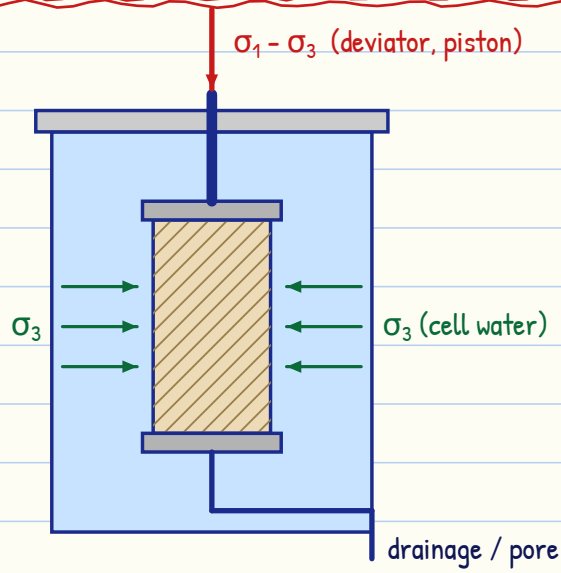
Advantages	Disadvantages
simple, quick and cheap equipment	the failure plane is forced (not the weakest plane)
easy to prepare the specimen, good for sands	non-uniform stress distribution, stress concentration at the edges
thin specimen: quick drainage and consolidation	no control on drainage; pore pressure not measured (only the total stress)
the residual strength can be found by reversal of the box	area of the shear plane changes during the test

Example 3

Direct shear on a dry sand (60 x 60 mm box) gives shear forces at failure of 208, 417 and 625 N for the normal loads of 360, 720 and 1080 N.

- Area = 36 cm^2 = 3.6 x 10^{-3} m^2. σ = 100, 200, 300 kN/m^2; τ_f = 57.8, 115.8, 173.6 kN/m^2.
- tan φ = 173.6/300 = 0.579 so φ = 30°, and the intercept c = 0 (dry sand).

6. Triaxial Compression Test (IS 2720 Part 11, 12)



Cylindrical specimen, L / D = 2
(38 mm x 76 mm, or 50 x 100 mm)
Rubber membrane, porous stones,
cell fluid gives all round σ_3.

Axial load raises σ_1 until failure
$\sigma_1 = \sigma_3$ + (deviator stress)

The most reliable and versatile test: the stress conditions are controlled, drainage can be controlled and the pore pressure measured. The specimen is surrounded by a fluid of pressure σ_3 (cell pressure), then the vertical stress is increased by the axial load through the piston up to failure: $\sigma_1 = \sigma_3 + \sigma_d$.

Procedure

- The specimen (38 mm dia x 76 mm length, or 50 x 100 mm; L/D = 2) is covered with a rubber membrane, put between the porous stones (filter paper strip along the side for quicker drainage), sealed with O-rings on the pedestal and the top cap and enclosed in

the cell filled with water.

- Cell pressure σ_3 is applied (and the specimen is saturated by back pressure if required). The axial strain rate is kept at 0.5 - 1.25 % per minute (undrained) or lower for drained tests (so that excess pore pressure dissipates).
- The load (proving ring) and the deformation are noted until the peak deviator stress or 15 - 20 % axial strain. The cross-section area is corrected: $A = A_0 / (1 - \varepsilon)$.

$$(\sigma_1 - \sigma_3)_f = P / A, \quad A = A_0 / (1 - \varepsilon)$$

- The Mohr's circles for 3 values of σ_3 are drawn and the common tangent gives c and φ.

Advantages over the direct shear test

- Drainage conditions and the pore water pressure are controlled / measured; the specimen fails along the weakest plane; the stress distribution is uniform; the specimens can be saturated; the effective stress parameters can be obtained; the complete stress path is available.
- Disadvantages: costly and elaborate equipment, skilled operator, long testing time (CD test on clay may take days), and $\sigma_2 = \sigma_3$ (axi-symmetric) only.

Example 4

CD triaxial tests on a soil give $\sigma_1 = 473$ kN/m^2 when $\sigma_3 = 100$ kN/m^2 and $\sigma_1 = 773$ kN/m^2 when $\sigma_3 = 200$ kN/m^2. Find c and φ.

- Subtract: $300 = 100\ N_\varphi$ so $N_\varphi = 3$, $\tan(45^\circ + \varphi/2) = 1.732 \rightarrow 45^\circ + \varphi/2 = 60^\circ \rightarrow \varphi = 30^\circ$.
- $\sigma_1 = \sigma_3 N_\varphi + 2c\sqrt{N_\varphi}$: $473 = 300 + 2c(1.732) \rightarrow c = 173/3.464 = 50$ kN/m^2.

7. Unconfined Compression Test (IS 2720 Part 10)

A special case of the UU triaxial test with $\sigma_3 = 0$. The cylindrical specimen (38 mm x 76 mm) of saturated cohesive soil is loaded axially in a compression machine (strain 1.25 %/min) until failure. The unconfined compressive strength q_u is the maximum axial stress (or the stress at 20 % strain).

- Failure circle passes through the origin: $\sigma_3 = 0$, $\sigma_1 = q_u$; as $\varphi_u = 0$ for the saturated clay, the envelope is horizontal.

$$c_u = q_u / 2 \quad (= \text{undrained shear strength } s_u), \quad \text{failure plane at } 45^\circ$$

- The test is used only for saturated clays (it can not be used for sands, dry soils, fissured clays or partially saturated soils, as they cannot stand without confinement).
- Consistency by q_u: very soft < 25 kPa, soft 25 - 50, medium 50 - 100, stiff 100 - 200, very stiff 200 - 400, hard > 400 kPa.

Sensitivity

S_t = (q_u undisturbed) / (q_u remoulded) at the same water content

- S_t = 1 - 2 insensitive; 2 - 4 normal; 4 - 8 sensitive; 8 - 16 extra sensitive; > 16 quick clay. Loss of strength on remoulding is due to the destruction of the structure; thixotropy is the regain of the strength with time at constant water content (e.g. piles driven in clay).

Example 5

An undisturbed clay specimen has q_u = 120 kN/m^2; remoulded at the same water content q_u = 30 kN/m^2.

- c_u = 120/2 = 60 kN/m^2; S_t = 120/30 = 4 (sensitive clay).

8. Vane Shear Test (IS 2720 Part 30)

For very soft to medium clays and sensitive clays which are difficult to sample; the in-situ test is done in the field by pushing a four-bladed vane (d = 75 mm, h = 150 mm; h = 2d) to the required depth and rotating it at 0.1°/s (6° per min) by a torque head.

- The soil is sheared along a cylindrical surface of the diameter d and height h, and the end circular areas. Torque at failure T = (resistance of the cylindrical surface) + (resistance of two ends):

$$T = \pi d^2 h s/2 + \pi d^3 s/6 \quad \rightarrow \quad s = T / [\, \pi (d^2 h/2 + d^3/6) \,]$$

- For the vane with only the top end in shear: $d^3/12$ instead of $d^3/6$. The vane shear gives the undrained strength (c_u) of the clay ($s = c_u$); the residual (remoulded) strength is found by rotating the vane rapidly several times.
- Advantages: quick, cheap, in-situ, the least disturbance, gives the sensitivity.
  Limitations: not suitable for stiff clays, sands and gravels; the strength is anisotropic; the rate effect and the vane insertion disturbance.

Example 6

Vane: d = 75 mm, h = 150 mm; the torque at failure T = 40 N-m.

- $d^2h/2 = (0.075)^2(0.15)/2 = 4.219 \times 10^{-4}$ m^3; $d^3/6 = 7.03 \times 10^{-5}$ m^3; sum = 4.922×10^{-4}.
- $s = 40 / (\pi \times 4.922 \times 10^{-4}) = 40/1.546 \times 10^{-3} = 25.9$ kN/m^2. So c_u = 25.9 kPa (soft clay).

9. Strength Tests Based on Drainage Conditions

The strength of a clay is not a constant: it depends on the drainage during loading. Three tests are carried out in the triaxial equipment, depending on the field conditions:

Test	Stage 1 (σ_3 applied)	Stage 2 (shear)	Result
UU Unconsolidated Undrained (Quick)	drainage closed, no consolidation	drainage closed	total stress: c_u, φ_u
CU Consolidated Undrained	drainage open, consolidates	drainage closed (u measured)	c_{cu}, φ_{cu}; c', φ' with u
CD Consolidated Drained (Slow)	drainage open, consolidates	drainage open, slow shear (u = 0)	c_d, $\varphi_d = c'$, φ'

(a) UU test

- Saturated clay: an increase of the cell pressure is taken up by the pore water entirely (B = 1), so the effective stress does not change, and the strength is independent of σ_3: all the circles have the same diameter and the envelope is horizontal:

$$\varphi_u = 0, \quad c_u = (\sigma_1 - \sigma_3)_f / 2$$

- Used for: end of construction stability of an embankment on soft clay, quick loading of a footing on clay, where the clay has no time to drain (short term - total stress analysis).
- For the partially saturated soil $\varphi_u > 0$ (the air compresses).

(b) CU test

- Specimen consolidated under σ_3 (drainage open), then sheared undrained. The pore pressure u is measured, hence the effective stress circles can also be drawn: $\sigma_1' = \sigma_1 - u_f$, $\sigma_3' = \sigma_3 - u_f$. The effective envelope gives c', φ' ($\varphi' > \varphi_{cu}$).
- Used for: rapid draw down of the earth dams, loading after the clay has consolidated (the next stages of construction), the strength gain with consolidation.

(c) CD test

- The soil is consolidated and then sheared so slowly that no excess pore pressure develops: total stress = effective stress. It gives the effective stress parameters c_d (c' ~ 0 for NC clay) and φ_d.

- Used for: long-term stability (the steady state after many years - cuts, natural slopes, retaining wall backfill), the sands (always drained), and the design of foundations on free-draining soils. The test takes a long time in clay.

Pore pressure parameters (Skempton)

$$\Delta u = B\,[\,\Delta\sigma_3 + A\,(\Delta\sigma_1 - \Delta\sigma_3)\,]$$

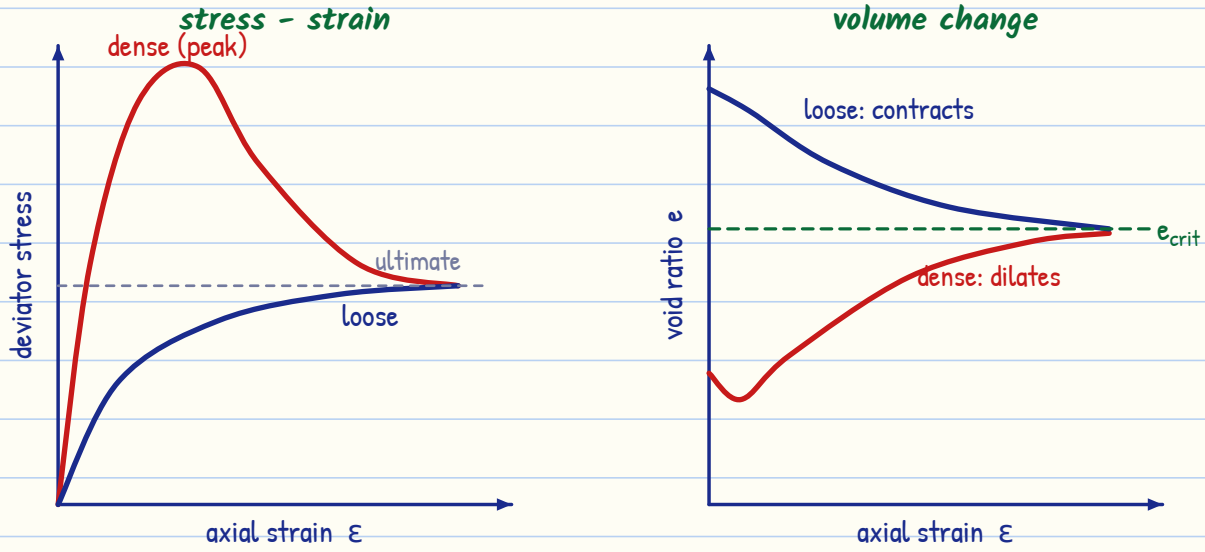
- B = $\Delta u / \Delta\sigma_3$ for the isotropic loading: B = 1 for the saturated soil, B = 0 for dry soil (the check of the saturation, B ≥ 0.95).
- A = the pore pressure parameter at the shearing: A_f = 0.5 - 1.0 normally consolidated clays, 0.25 - 0.5 lightly OC clay, 0 to -0.5 heavily OC clay (negative u due to dilation), 1 - 3 for sensitive clay. Dense sand A is negative.

Example 7

A saturated clay in the CU test: σ_3 = 150 kN/m^2, the deviator stress at failure = 130 kN/m^2, pore pressure at failure u_f = 70 kN/m^2. Find the effective principal stresses and A_f.

- σ_1 = 150 + 130 = 280; σ_1' = 280 - 70 = 210 kN/m^2; σ_3' = 150 - 70 = 80 kN/m^2.
- With B = 1: 70 = 1[0 + A(130)] -> A_f = 0.54; sin φ' = (210-80)/(210+80) = 0.448 so φ' = 26.6°.

10. Shear Behaviour of Sands - Critical Void Ratio



Dense and loose sand

- Dense sand (D_r high, low void ratio): the stress rises to a sharp peak at small strain, then falls to an ultimate value. The volume first decreases slightly, then increases (dilatancy) - the grains have to ride over one another (interlocking). Pore pressure in the undrained shear is negative.
- Loose sand: the stress rises slowly to the ultimate value without a peak and the volume decreases (contraction) all through. In undrained shear the pore pressure is positive.
- Both the dense and the loose sand, at large strain, reach the same ultimate (critical) strength and the same void ratio. φ_{peak} (dense) > φ_{ult} (loose): $\varphi_{peak} = \varphi_{cv}$ + dilation angle ψ.

Critical void ratio (Casagrande, 1936)

Critical void ratio e_c is the void ratio at which the sand undergoes no volume change during shearing, at a given confining pressure: a sand with $e > e_c$ (looser) contracts while at $e < e_c$ (denser) dilates.

- e_c is not a constant - it depends on the confining pressure: it decreases when the confining pressure (σ_3) increases. It is determined by a series of drained triaxial / direct shear tests on the sand specimens at different initial void ratios under the same σ_3: plot the volume change ($\Delta V/V$) at failure against the initial e; e at $\Delta V = 0$ is e_c.
- Significance: a saturated loose sand ($e > e_c$) with a tendency to contract under shear (or vibration) produces an increase in pore pressure, decrease in effective stress, and ultimately a flow failure (liquefaction). A dense sand ($e < e_c$) dilates and gains the strength. The critical density (relative density about 40 - 50 %) divides the two.
- The idea of critical void ratio has now developed to the critical state of soil mechanics: at large strain the soil flows at constant stress, constant volume.

II. Liquefaction

Liquefaction is the phenomenon in which a saturated, loose, cohesionless soil loses its shear strength due to a sudden rise in the pore water pressure under the dynamic (cyclic) loading, such as an earthquake, and behaves like a viscous liquid.

Mechanism

- Shaking (or other fast loading) tends to compact the loose sand, but the water cannot drain out in the short time (undrained condition): the load is transferred to the pore water, the excess pore pressure builds up with each cycle.
- Effective stress $\sigma' = \sigma - u$. When the excess pore pressure becomes equal to the total overburden stress ($u = \sigma$), $\sigma' = 0$ and, since $\tau_f = \sigma' \tan \varphi = 0$, the soil has no shear strength and flows. Pore pressure ratio $r_u = u/\sigma'_0 = 1$.

Strength: $\tau_f = (\sigma - u) \tan \varphi \rightarrow 0$ as $u \rightarrow \sigma$

- The excess water moves upward and appears at the surface as sand boils (mud volcanoes); the ground sinks and large flows take place.

Conditions necessary (factors affecting)

Factor	Liquefaction is likely when
Soil type	uniform fine to medium sand, non-plastic silt (poorly graded, $C_u < 2 - 4$); not in clays or in gravels
Density	loose, $D_r < 50$ %, SPT N-value < 15 (corrected $(N_1)_{60} < 30$ for the sandy soils)
Saturation	below the ground water table, usually within the upper 15 - 20 m
Drainage	poor drainage, so the pore pressure cannot dissipate during loading
Earthquake	high intensity (magnitude > 5), long duration (many cycles), peak ground acceleration > 0.1 g
Confining pressure	low effective overburden stress (shallow depth)

Types

- Flow liquefaction: loose soil, static shear stress greater than the residual strength; large displacements (flow slides) - Lower San Fernando dam, 1971.
- Cyclic mobility: limited, progressive deformation of the medium-dense soils during cyclic loading; lateral spreading near water fronts.

Effects and examples

- Bearing failure (tilting and sinking of buildings: Niigata, Japan, 1964, apartment buildings tilted), floating up of buried tanks, manholes and pipelines, settlement, lateral spreading of the slopes and river banks, failure of the earth dams, and damage to the retaining walls and ports (Kobe 1995). In India: Bhuj (Gujarat) earthquake 2001, Bihar - Nepal 1934, Assam 1950.

Evaluation

- Simplified procedure (Seed and Idriss): factor of safety F.S. = CRR / CSR against liquefaction, where CSR (cyclic stress ratio) = $0.65 (a_{max}/g)(\sigma_v/\sigma_v') r_d$ is the demand, CRR (cyclic resistance ratio) is the capacity obtained from the SPT / CPT / lab cyclic

triaxial tests. F.S. < 1 (generally < 1.2 – 1.3) shows liquefaction.

Mitigation measures

- Densification: vibro flotation, dynamic compaction, compaction piles, blasting.
- Drainage: gravel (stone) columns, earthquake drains – quick dissipation of the pore pressure; lowering the water table.
- Grouting / chemical stabilisation, soil replacement; reinforcing: deep foundations (piles) through the liquefiable layer, mat foundations, structural measures.

12. Shear Strength of Clays – Other Points

- Normally consolidated (NC) clay: $c' = 0$, the envelope passes through the origin, $\varphi' = 20^\circ$ – 30° (reduces as the PI increases). Over consolidated (OC) clay: the envelope is curved up to the preconsolidation pressure, $c' > 0$ for small stresses, and the strength is much more than that of the NC clay at the same stress.
- Peak and residual strength: for stiff OC clays, the strength is reduced to the residual strength ($c_r \sim 0$, $\varphi_r < \varphi'$) after large displacement – used in the analysis of old landslides (ring shear test).
- Strength of the partially saturated soil: the suction (negative pore pressure) gives the extra apparent cohesion; vanishes on saturation (slope failure after rains).
- Factors affecting the shear strength: soil type, particle size, shape and gradation, density (void ratio), water content, drainage conditions, effective stress, stress history, rate of loading, structure and cementation, temperature.

Stress path

- The path of the point $(p, q) = ((\sigma_1+\sigma_3)/2, (\sigma_1-\sigma_3)/2)$ during the loading. The K_f line has the slope $\tan\alpha = \sin\varphi'$ and the intercept $a = c' \cos\varphi'$. The effective stress path of the NC clay in the CU test bends to the left, of the OC clay to the right.

13. Quick Revision

Item	Formula
Mohr - Coulomb	$\tau_f = c + \sigma \tan \varphi$; $\tau_f = c' + \sigma' \tan \varphi'$
Plane at θ (major plane)	$\sigma_n = (\sigma_1+\sigma_3)/2 + (\sigma_1-\sigma_3)/2 \cos 2\theta$, $\tau = (\sigma_1-\sigma_3)/2 \sin 2\theta$
Failure plane	$\theta_f = 45^\circ + \varphi/2$
Principal stresses at failure	$\sigma_1 = \sigma_3 \tan^2(45^\circ+\varphi/2) + 2c \tan(45^\circ+\varphi/2)$
For sand	$\sin \varphi = (\sigma_1-\sigma_3)/(\sigma_1+\sigma_3)$
UCC	$c_u = q_u/2$; $S_t = q_{u(und)} / q_{u(rem)}$
Vane shear	$s = T / [\pi (d^2h/2 + d^3/6)]$
Skempton	$\Delta u = B [\Delta\sigma_3 + A (\Delta\sigma_1 - \Delta\sigma_3)]$
Effective stress	$\sigma' = \sigma - u$; at liquefaction $u = \sigma$, $\sigma' = 0$
Liquefaction check	F.S. = CRR / CSR

Important questions (exam)

- Explain the importance of shear strength of soil. State and explain the Mohr - Coulomb theory of failure and show that $\theta = 45^\circ + \varphi/2$.
- Derive the relation between the major and minor principal stresses at failure in terms of c and φ.
- Describe the direct shear test: the procedure, merits and demerits. How are c and φ found?
- Explain the triaxial test. Differentiate between UU, CU and CD tests; state where each is used.
- What is the unconfined compression test? Define the sensitivity of clay. Explain the vane shear test.
- What is the critical void ratio? How does it affect the behaviour of sand during shear?
- What is liquefaction? Explain the factors that cause it and the measures to prevent it.
- Numerical: Mohr's circle problems, c and φ from the triaxial / direct shear data, c_u from q_u, vane shear.